TAED: Temporally-Aware Efficient Diffusion using Reinforcement Learning

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Abstract

Denoising Diffusion Probabilistic Models (DDPMs) offer high-fidelity image synthesis but suffer from slow inference due to numerous sequential steps. Existing accelerators like DDIM often use fixed schedules. We propose Temporally-Aware Efficient Diffusion (TAED), utilizing Reinforcement Learning (RL) for adaptive timestep skipping during reverse diffusion to reduce computation. Formulating this as a sequential decision task, an RL agent learns a dynamic skipping policy. We explored various RL algorithms (REINFORCE, A2C, PPO), policy architectures (LSTM/MLP), and reward designs (sparse terminal, dense MSE vs. reference, dense perceptual using VGG/noise metrics). On MNIST, TAED achieved decent step reductions. However, maintaining high final image fidelity proved challenging across configurations, often yielding noisy results. This highlights fundamental difficulties in designing step-wise rewards that correlate with global perceptual quality when skipping steps adaptively. Reward engineering emerges as the central obstacle. Future work should investigate hybrid rewards combining sparse final perceptual metrics (e.g., LPIPS) with dense guidance, and integration with deterministic samplers like DDIMs.

1. Introduction

Denoising Diffusion Probabilistic Models (DDPMs) [2] have established themselves as the state-of-the-art in generative modeling, particularly for producing diverse and high-fidelity images. Their core mechanism, however, relies on an iterative reverse diffusion process. This process typically necessitates hundreds or thousands of sequential function evaluations, commonly employing a U-Net backbone [7], to synthesize a single sample. The resulting computational expense and high inference latency significantly curtail the applicability of DDPMs in scenarios demanding real-time performance or operating under resource constraints.

Significant research efforts have focused on mitigating this computational bottleneck by accelerating diffusion model inference. Techniques such as Denoising Diffusion Implicit Models (DDIMs) [11] enable sampling with substantially fewer steps by facilitating larger, non-Markovian jumps along the reverse trajectory. Nonetheless, the predominant acceleration strategies, including DDIMs, typically operate based on predefined, **fixed** step schedules. These schedules are applied uniformly across all generated samples, irrespective of their intrinsic complexity or the state of the generative process at intermediate stages.

A critical observation motivating dynamic approaches, noted by [11] among others, is the non-uniform contribution of individual denoising steps to the final image quality. Early steps in the reverse process predominantly reconstruct coarse features from near-Gaussian noise, whereas later steps are primarily responsible for refining high-frequency details and subtle textures. Consequently, adhering to a fixed schedule may lead to redundant computations on steps offering minimal perceptual improvement, thereby neglecting opportunities for instance-specific optimization.

In this work, we investigate an alternative acceleration paradigm: **adaptive timestep skipping**, guided by Reinforcement Learning (RL). Instead of employing a fixed schedule, we propose training an RL agent to dynamically determine at each potential step whether to *execute* denoising operation or to *skip* it, conditioned on the current state (intermediate sample, hidden state and timestep). This formulation recasts the accelerated sampling procedure as a Markov Decision Process (MDP), wherein the agent learns a policy $\pi(a_t|s_t)$ aimed at maximizing a cumulative reward signal. A central challenge and focus of our investigation lies in the **design of this reward function**, exploring various strategies to effectively balance computational efficiency (step reduction) with the preservation of final image fidelity.

Our primary contributions are articulated as follows:

- The formulation of adaptive diffusion timestep skipping as a sequential decision-making problem solvable via Reinforcement Learning.
- An investigation into reward function design, encompassing sparse terminal rewards, dense rewards based on Mean Squared Error (MSE) penalizing deviations

from a reference trajectory, and dense rewards engineered to promote step-wise perceptual progress.
- An analysis of the distinct skipping behaviors learned under different reward schemes and algorithmic choices, critically identifying the inherent difficulties in designing effective dense rewards that reliably correlate with global perceptual quality when adaptive skipping actions are permitted.

Our experiments, conducted primarily on the MNIST dataset, demonstrate that RL agents can successfully learn policies achieving significant step-saving rates. However, consistently generating high-fidelity images under these adaptive schedules proved challenging across the explored configurations, underscoring that effective reward engineering remains a primary obstacle for RL-based adaptive diffusion acceleration.

2. Related Work

2.1. Denoising Diffusion Implicit Models (DDIMs)

DDIMs [11] have emerged as a significant advancement in diffusion-based generative modelling, offering deterministic sampling and accelerated inference compared to DDPMs [2]. The key innovation of DDIMs lies in their ability to formulate the reverse diffusion process as a non-Markovian process, enabling the use of fewer sampling steps without compromising sample quality. This deterministic nature allows for various applications, including latent space interpolation and editing.

Several works have explored the properties and applications of DDIMs. For instance, researchers have investigated the impact of different sampling schedules on the trade-off between speed and quality [8]. In addition, efforts have been made to further accelerate DDIM inference through techniques such as knowledge distillation and model pruning [4]. However, the fixed-step schedule in standard DDIM inference can still lead to redundant computations, motivating the need for adaptive sampling strategies. Our work extends these efforts by introducing an RNN-based approach to dynamically optimize the DDIM sampling trajectory, enabling intelligent step skipping.

2.2. AdaDiff

A recent approach, "AdaDiff: Adaptive Step Selection for Fast Diffusion Models" [14], moves beyond one-size-fits-all strategies by conditioning on input text's richness. They introduce a lightweight framework which learns instance-specific step usage policies. This is achieved through a policy gradient method that maximizes a reward function by balancing inference time and generation quality. The key idea is that the number of denoising steps should be tailored to each input text, not a fixed value, allowing for more efficient generation when the input text is less complex. AdaDiff demonstrates strong results on both image and video generation benchmarks, achieving similar visual quality compared to the baseline using a fixed 50 denoising steps while reducing inference time by at least 33%, going as high as 40%. Furthermore, AdaDiff can be combined with other acceleration techniques to improve speed.

Our work builds upon the concept of adaptive sampling by introducing a Reinforcement Learning (RL)-based framework that learns to dynamically skip denoising steps during the diffusion process. Rather than following a fixed or pre-defined schedule, our method trains a policy network to decide, at each timestep, whether to perform a denoising operation or skip it, based on the current image state and temporal context. This enables instance-specific acceleration while maintaining generation quality.

2.3. Denoising Diffusion Step-Aware Models (DDSM)

Our inspiration to look for step-wise redundancies in the generation process comes from this work [13]. It discusses the inefficiencies in standard denoising diffusion probabilistic models (DDPMs), where a fixed network is used for all denoising steps, leading to excessive computational costs. The authors propose Denoising Diffusion Step-Aware Models (DDSM), which dynamically allocate computational resources across different steps by utilizing a slimmable network and evolutionary search to optimize efficiency without sacrificing generation quality.

DDSM builds upon previous work in diffusion model acceleration, which has primarily focused on reducing the number of denoising steps. Notable approaches include Denoising Diffusion Implicit Models (DDIM) [11], which introduce a non-Markovian sampling process to accelerate inference by skipping redundant steps, and Pseudo Numerical Methods for Diffusion Models (PNDM) [3], which refine step selection through numerical solvers. These methods aim to decrease the number of function evaluations but still use a fixed model architecture across all timesteps.

Overall, DDSM advances diffusion model acceleration by integrating adaptive network architectures with step-aware scheduling. In a way, DDSM proves that some steps requires "less" computation power or less importance in the process of denoising in generation process. This give us motivation for our novel approach "step skipping" using RNNs.

3. Methodology: Adaptive Skipping via RL

Diffusion models have demonstrated exceptional generative quality but are notoriously computationally expensive during inference due to the large number of denoising steps (typically 1000 steps). A key observation is that not all timesteps contribute equally to the final image quality. This calls for an **adaptive skipping strategy**, where a policy can learn what steps to skip based on the current context. Since we don't have a proper supervision to learn to skip a step, we will use try to use RL by modelling the problem as a MDP (**Markov Decision Process**).

In simple terms, an agent observes the current state of the generation process and decides whether to perform the computationally expensive denoising step or skip it.

3.1. MDP Formulation

We define the generation process as a Markov Decision Process (MDP) $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$:

- **State ($\mathcal{S}$):** $s_t = (x_t, t, h_t)$, including the current noisy image $x_t \in \mathbb{R}^{C\times H\times W}$, the reverse timestep $t \in \{T-1, ..., 0\}$, and a recurrent hidden state h_t of LSTM which captures the temporal dependencies.
- **Action ($\mathcal{A}$):** $a_t \in \{0, 1\}$.
- **Transition ($\mathcal{P}$):** $x_{t-1} = D(x_t, t)$ if $a_t = 0$, else $x_{t-1} = x_t$, where $D(x_t, t)$ is one diffusion denoising step. The hidden state transition $h_{t-1} = f(h_t, x_t, t)$ depends on the policy network. So the new state, $s_{t-1} = (x_{t-1}, t-1, h_{t-1})$.
- **Reward ($\mathcal{R}$):** $r_t = R(s_t, a_t)$, step-wise reward. More on this in 3.3
- **Discount Factor (γ):** Typically close to 1 , used to bound the return of the trajectory. We have used 0.99 in our experiments

3.2. Policy and Value Network Architectures

The RL agent's decision-making relies on neural networks approximating the policy and, for actor-critic methods, the value function.

Policy Network (π_θ): The policy network, $\pi_\theta(a_t|s_t)$, is central to the RL agent. Its function is to map the current state s_t to a probability distribution over the possible actions (denoise or skip). Specifically, for our binary action space, it outputs the probability $p_t = \pi_\theta(a_t = 1|s_t)$ of taking the 'skip' action at timestep t.

Given the state $s_t = (x_t, t, h_t)$ contains spatial image data (x_t) and temporal information (t, h_t), suitable architectures must process both modalities. Although architectures like ConvLSTMs, which directly handle spatial sequences, can be considered, their significant computational cost per step presents a challenge, particularly during the training phase involving multiple network evaluations. Since the data we are handling are gray scale 28×28 images, we choose the following architecture,

Implemented Policy and Value Network Architecture: The overall model consists of a shared feature encoder, a recurrent decision module for temporal processing, and two separate output heads: one for predicting the skip probability (policy) and another for estimating the state value (critic), as illustrated in Figure 1.

1. **Image Feature Extraction:** The input image $x_t \in \mathbb{R}^{C\times H\times W}$ is processed through a convolutional feature extractor to obtain a compact visual representation. This step reduces spatial complexity while retaining essential local structures required for decision making.
2. **Time Embedding:** The current timestep t is normalized and embedded into a fixed-size vector through a learnable projection. This embedding captures the temporal context of the diffusion process.
3. **Feature Combination:** The visual features and the time embedding are concatenated to form a unified feature vector that serves as input to the recurrent policy network.
4. **Sequential Processing (LSTM):** The combined features are passed through a recurrent module (LSTM), which maintains temporal state and captures history-dependent information necessary for adaptive timestep decisions. The LSTM produces updated hidden and cell states (h_t, c_t) along with an output representation.
5. **Skip Head (Policy):** The LSTM output is passed through a projection head that computes the skip probability $p_t \in [0, 1]$ via a sigmoid activation. This value represents the agent's confidence in skipping the current denoising step.
6. **Value Head (for Actor-Critic):** Simultaneously, the LSTM output is processed through a parallel projection head to estimate the state value $V_\phi(s_t)$. This output is used for computing advantage estimates in actor-critic algorithms such as A2C and PPO.

The architecture is designed to be modular and adaptable, allowing end-to-end training of both policy and value functions while enabling flexible design choices for the convolutional encoder and projection layers.

3.3. Reward Function Design Exploration

The design of the reward function is arguably the most critical aspect of applying RL, as it directly shapes the agent's learned behavior. In general RL, the reward signal r_t provided after taking action a_t in state s_t should ideally "quantify" the immediate "goodness" of that action. The agent's objective is typically to maximize the expected cumulative discounted reward over a trajectory. Therefore, rewards should incentivize actions that

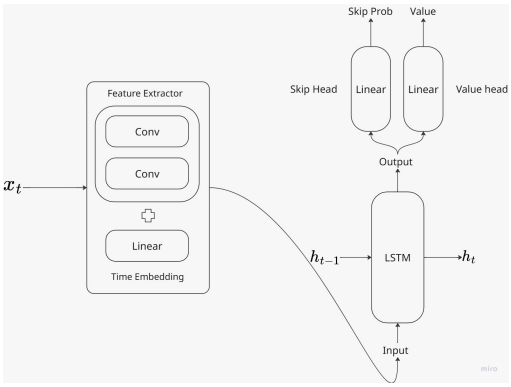


Figure 1. Architecture of the TAED policy and value network. The input image is encoded via a convolutional feature extractor and time embedding, then passed to an LSTM for sequential processing. The output is fed to two heads: one for computing the skip probability (policy), and another for estimating the state value (critic).

lead towards desirable final outcomes.

In our adaptive skipping task, the overall goal is twofold: achieve high final image quality while minimizing the number of computationally expensive denoising steps. This translates into designing a reward signal that encourages the agent to,

- Take the '**denoise**' action ($a_t = 0$) when the step is crucial for improving or maintaining the eventual quality of the image x_0.
- Take the '**skip**' action ($a_t = 1$) when the step is less critical, thereby saving computation, provided this skip does not excessively effect the final image quality.

We can conceptualize this via two components: $R_{quality}$, reflecting the impact on image fidelity, and R_{speed}, reflecting computational efficiency. The challenge lies in effectively combining these, possibly conflicting, objectives into a step-wise or terminal reward signal r_t or R_{final} that yields the desired final behavior. We explored several strategies with varying levels of success.

3.3.1 Strategy 1: Sparse Terminal Rewards

The most straightforward approach assigns zero reward for all intermediate steps ($r_t = 0$ for $t > 0$) and provides a single reward R_{final} at the end of the trajectory based on the final state x_0. This terminal reward combined the two objective,

$$R_{final} = \alpha \cdot R_{quality}(x_0, x_{ref_0}) + (1-\alpha) \cdot R_{speed}(S, T). \quad (1)$$

where,

- $R_{\text{quality}} = \lambda_{\text{vgg}} \cdot R_{\text{vgg}} + \lambda_{\text{KL}} \cdot R_{\text{KL}}$
- $R_{\text{speed}} = 1 - \frac{S}{T}$
- R_{vgg} : Uses the negative Mean Squared Error (MSE) between intermediate VGG16 features of generated and reference images to reward the perceptual similarity.
- R_{KL} : Quantifies distribution similarity via the negative Kullback-Leibler (KL) divergence between normalized pixel distributions of generated and reference images.

Challenges Encountered: While simple, this sparse reward structure suffered immensely from adaptive skipping problem. With a reward signal only after $T = 1000$ steps, the agent struggled to determine which specific skip/denoise actions early in the trajectory contributed positively or negatively to the final outcome. This led to very high variance gradients, unstable training, and ultimately converged to policies yielding poor quality final images (high FID), as reported in the mid-term report.

3.3.2 Strategy 2: Dense Rewards Penalizing Deviation

To provide more immediate feedback, we implemented dense step-wise rewards r_t calculated at each timestep t. This approach utilized a pre-computed reference trajectory $\{x_{ref_k}\}_{k=0}^{T}$ generated without skipping.

- If $a_t = 0$, (Denoise) agent state becomes x'_{t-1}:

$$r_t = -w_q \cdot \text{MSE}(x'_{t-1}, x_{ref_{t-1}})$$

- If $a_t = 1$ (Skip), agent state remains x_t:

$$r_t = -w_p \cdot \text{MSE}(x_t, x_{ref_{t-1}}) + skip_bonus$$

Here, $w_q, w_p > 0$ weight the MSE penalties against the reference target state x_{ref_t-1}, and $skip_bonus > 0$ is given to cater the speed in computation.

Challenges Encountered:

- **Hyperparameter Tuning Difficulty:** Balancing the penalty weights (w_q, w_p) and the skip reward ($skip_bonus$) was crucial yet challenging.
- **Low Penalty Weights:** Resulted in agents prioritizing the $skip_bonus$, leading to excessive skipping ($\approx$75% steps saved) and poor final image quality (noise).
- **High Penalty Weights:** This encouraged skipping only if necessary as you can see in Figure 2. But **failed** to guarantee **perceptual quality**, as optimizing step-wise MSE against a fixed reference is not equivalent to optimizing quality under adaptive skipping.

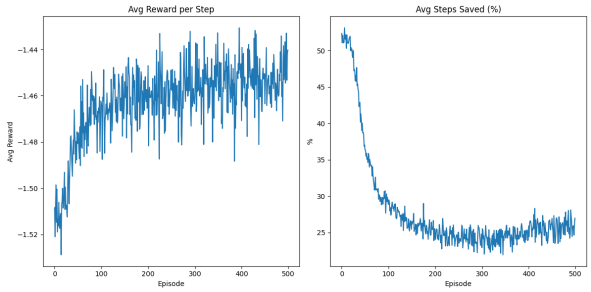


Figure 2. Strategy 2 : Rewards & Skip steps

3.3.3 Strategy 3: Dense Rewards for Perceptual and Noise Reduction

This strategy provides dense, step-wise rewards to guide the policy based on perceptual improvement, noise reduction, and computational efficiency.

At timestep t, the reward is:

$$r_t = \alpha\, r_{\text{quality}} + \beta\, r_{\text{noise}} + \gamma_{\text{cost}}\, r_{\text{cost}} \tag{2}$$

where α , β and γ_{cost} are manually tuned weight parameters and $r_{\text{quality}}, r_{\text{noise}}$ and r_{cost} are the corresponding rewards discussed below

Perceptual improvement. Let $F(\cdot)$ denote the VGG feature extractor, and x_{ref} the final reference image. This term measures the reduction in perceptual error before and after denoising, applied only when the agent chooses to denoise ($a_{t,\text{final}} = 0$):

$$\begin{aligned} r_{\text{quality}} = \big[&\text{MSE}\big(F(x_t), F(x_{\text{ref}})\big) \\ &- \text{MSE}\big(F(x'_{t-1}), F(x_{\text{ref}})\big)\big] \cdot (1 - a_{t,\text{final}}) \end{aligned} \tag{3}$$

Here, x_t is the current state, and x'_{t-1} is the denoised version of x_t.

Noise reduction. This term encourages steps that reduce predicted noise magnitude:

$$r_{\text{noise}} = \|\hat{\epsilon}_{t+1}\|_2^2 - \|\hat{\epsilon}_t\|_2^2 \tag{4}$$

where $\hat{\epsilon}_t$ is the predicted noise at step t from the diffusion model.

Computation penalty. A fixed cost is applied when the agent performs a denoising step:

$$r_{\text{cost}} = -(1 - a_{t,\text{final}}) \tag{5}$$

The weights α, β, and $\gamma_{\text{cost}} > 0$ determine the influence of each component. This dense reward formulation offers continuous guidance and balances quality, denoising progress, and compute cost. However, it introduces extra overhead due to per-step VGG feature extraction.

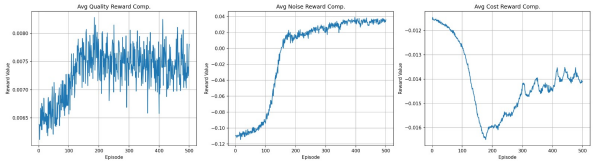


Figure 3. Strategy 3: Visualising various rewards over the training phase

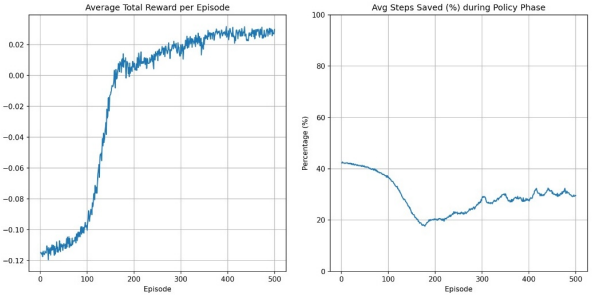


Figure 4. Strategy 3: Rewards & Skip steps

3.4. RL Training Algorithms

We have used the following RL algorithms in our experimentation,

- **REINFORCE** [12]: Basic policy gradient method, known for high variance. Used with entropy regularization $H(\pi(a_t|s_t))$ added to the objective.
- **A2C (Advantage Actor-Critic)** [6]: Incorporates a learned value function $V_\phi(s_t)$ to estimate advantages $A_t = R_t - V_\phi(s_t)$, reducing variance. Updates use policy loss $\propto -\log \pi_\theta(a_t|s_t)A_t$ and value loss (e.g., MSE on $V_\phi(s_t)$ vs. target return R_t).
- **PPO (Proximal Policy Optimization)** [9]: Further stabilizes Actor-Critic by using a clipped surrogate objective function on the probability ratio $\rho_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$, constraining policy updates. Uses Generalized Advantage Estimation (GAE) [10] for lower variance advantage estimates. Allows multiple optimization epochs per collected batch.

4. Experiments

4.1. Experimental Setup

As mentioned in the section 3.3, we have designed each rewarding strategy and trained on RL algorithms mentioned in 3.4.

- Strategy 1 : Uses the 3.3.1 and we have trained reinforce agent to design skipping strategy.

- Strategy 2 : Uses the 3.3.2 and we have trained PPO agent to design the skipping strategy.
- **Strategy 3 (Our Final Approach)**: Uses the 3.3.3 and we have trained A2C agent to design the skipping strategy.

In particular, we experiment with two variants of the strategy 3. The **first variant** allows **unrestricted skipping**, relying solely on the learned reward signals to guide the policy. The **second variant** introduces a constraint on the number of consecutive skip actions using a `max_consecutive_skips` parameter, which forces a denoising step if the agent attempts to skip for too many steps in a row. Later we see that, this constraint improves training stability by ensuring the state remains within a denoising-recoverable trajectory, especially during early policy exploration.

4.2. Baselines for Comparison

We compare our models with the following baseline methods:

- **Standard Diffusion Model:** A conventional diffusion-based approach without any skipping or temporal awareness.
- **Random Skip:** At each timestep, we will skip randomly with a probability of 0.5.
- **Alternate Skip:** As the name suggests, we will skip every alternative step in the denoising process.
- **First 100 Steps:** In this skipping strategy we will skip first 100 denoising steps.
- **Last 100 Steps:** In this skipping strategy, we will skip last 100 denoising steps.

4.3. Evaluation Metrics

To evaluate the quality of generated images, we employ the standard metrics, Fréchet Inception Distance (FID) [1], Natural Image Quality Evaluator (NIQE) [5].

Fréchet Inception Distance (FID): FID measures the distance between feature distributions of real and generated images, computed using activations from the Inception-v3 network. Given two multivariate Gaussians with means μ_r, μ_g and covariances Σ_r, Σ_g, corresponding to real and generated image features respectively, FID is defined as:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r\Sigma_g\right)^{\frac{1}{2}}\right) \tag{6}$$

To compute the FID score, features are extracted from both the generated image $x_{\text{generated}}$ (using skipping strategy) and x_{real} (from MNIST test dataset) using a pretrained Inception network. These features are then modeled as multivariate Gaussians, and the FID measures the distance between their distributions.

Table 1. Comparison of skipping strategies on MNIST (T=1000). Lower FID/NIQE/Time is better.

Method	FID ↓	NIQE ↓	Time (s) ↓
Full DDPM (1000 steps)	5.59	4.55	13.5
First-100 Skip	**5.50**	4.56	12.0
Last-100 Skip	162.65	8.07	12.1
Alternate Skip (500 steps)	256.15	6.47	6.64
Random Skip (p=0.5)	206.85	6.49	6.60
RL (Strategy 1)	295.56	6.87	7.62
RL (Strategy 2)	288.5	7.32	7.97
RL (Strategy 3-1)	130.90	4.94	8.19
RL (Strategy 3-2)	20.49	4.9	10.24

Natural Image Quality Evaluator (NIQE). NIQE is a no-reference image quality metric based on a statistical model of natural scene statistics. It fits a multivariate Gaussian to patches from pristine natural images and compares it with the distribution of patches from a test image. Lower NIQE values indicate better perceptual quality.

$$\text{NIQE} = D_{\text{MA}}(\mu_t, \Sigma_t; \mu_n, \Sigma_n)$$

where D_{MA} is the Mahalanobis-like distance between the mean (μ_t) and covariance (Σ_t) of features from the test image, and the corresponding parameters (μ_n, Σ_n) from natural pristine statistics.

5. Results and Discussion

5.1. Quantitative Comparison with Baselines

Table 1 Evaluating the different RL reward strategies revealed significant challenges in balancing efficiency and quality. Transitioning from the sparse terminal reward (Strategy 1) to dense rewards penalizing step-wise MSE deviation from a reference trajectory (Strategy 2) offered minimal benefit, as shown in Table 1. While intended to provide more frequent guidance, this strategy resulted in a negligible FID improvement (295.56 to 288.5) and actually yielded a worse NIQE score (6.87 to 7.32). This performance, significantly poorer than simple baselines like 'First-100 Skip', underscores the difficulty of using intermediate MSE against a non-skipping path as an effective proxy for final image quality when adaptive skipping is employed, aligning with the challenges discussed in previous sections .

A different dense reward approach (Strategy 3) focused on rewarding perceptual progress (using VGG features and noise reduction metrics) during denoising steps, assigning zero reward for skipping. An initial variant allowing unrestricted skipping (Strategy 3-1) showed marked improvement over the MSE-based Strategy 2, achieving considerably better FID (≈130.90)

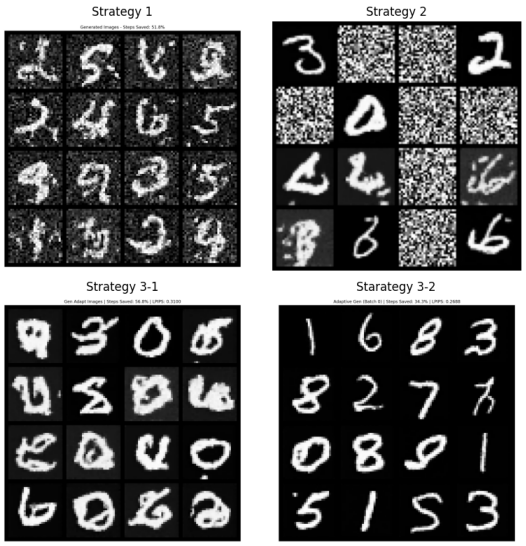


Figure 5. Generated MNIST samples. (a) Sparse Terminal Rewards. (b) Dense Rewards Penalizing Devia- tion. (c) Dense Rewards for Perceptual and Noise Reduction without max consecutive step limit (d) Dense Rewards for Perceptual and Noise Reduction with max consecutive step limit set to 2

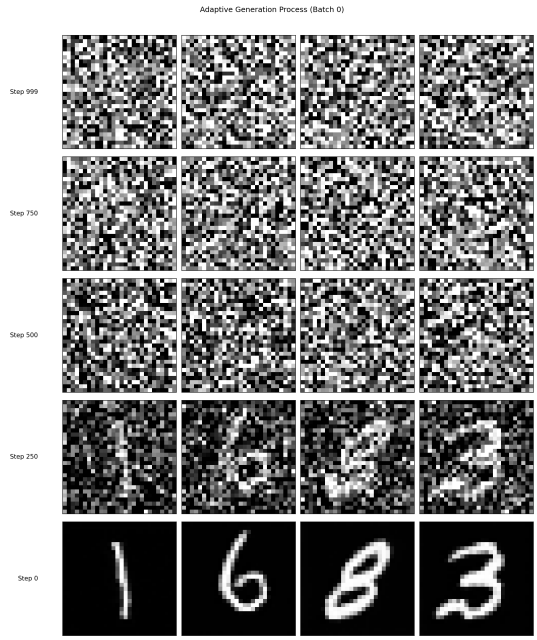


Figure 6. Progression of images over the steps based on strategy 3-2

and NIQE (4.94). This suggests incorporating perceptual information is beneficial. However, introducing a constraint limiting the maximum number of consecutive skips (Strategy 3-2) proved crucial. This constrained variant achieved the best results among all RL methods tested, with FID dropping substantially to ≈20.49 and a competitive NIQE of 4.90. This highlights that preventing excessive consecutive skips, thereby keeping the generation process closer to a stable trajectory, significantly impacts final quality, even when the reward directly incentivizes perceptual progress only during denoising steps. Despite this success relative to other RL approaches, Strategy 3-2 still fell short of the optimal FID achieved by non-adaptive methods like the full DDPM or 'First-100 Skip'.

5.2. Qualitative Analysis of Generated Samples

Visual inspection of generated samples, presented in Figure 5, confirms the challenges highlighted by the quantitative metrics. Strategy 1 (Sparse Reward) yields images barely recognizable as digits, dominated by noise, reflecting its poor FID score and the difficulty of learning from only a terminal reward (Fig. 5a). Strategy 2 (Dense MSE Penalty) shows mixed results (Fig. 5b); while some digit structures emerge, many samples collapse into noise or exhibit severe artifacts, indicating that minimizing step-wise MSE against a reference path does not reliably prevent catastrophic deviations when skipping. Strategy 3-1 (Dense Perceptual Progress, Unconstrained Skipping) produces more coherent digit shapes compared to Strategies 1 and 2, but samples suffer from distortion and lack fine details (Fig. 5c). Notably, Strategy 3-2 (Perceptual Progress with Consecutive Skip Constraint) generates significantly cleaner and more recognizable digits (Fig. 5d), aligning with its superior FID score. While still less sharp than samples from a full DDPM (not shown), these results underscore the positive impact of constraining skips for stabilizing the generation process under Strategy 3's reward scheme, leading to the best qualitative outcome among the explored RL methods.

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